Exercise 1: Entropy in Binary Classification

Suppose we are facing a binary classification learning problem with the feature space $\mathcal{X} = \{0, 1\}$ and label space $\mathcal{Y} = \{-1, 1\}$. Let us assume that the relationship between the feature X and the label Y is specified by

$$Y = 2 \max\{X, \epsilon\} - 1$$

where $\mathbb{P}_X = \text{Ber}(\theta_X)$, $\mathbb{P}_\epsilon = \text{Ber}(\theta_\epsilon)$, and X and ϵ are independent.

(a) What is $H(\epsilon)$?

(b) What is $H(Y|X)$?

(c) What is $H(Y, X)$?

(d) Now assume instead that $\epsilon = \psi(X)$ for some deterministic function $\psi : \{0, 1\} \to \{0, 1\}$, so X and ϵ are in general no longer independent. What are the answers to the previous questions in this case?